

Numerical Methods

Curve Fitting – Cubic Spline

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Piecewise Cubic Splines

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Cubic Spline

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Suitability

- It is very important to ensure the smoothness up to order 2 for interpolation needed in CAD/CAM, computer graphic, and robot path/trajectory planning. 平滑
- To constructing cubic functions $S(x)$ 三次函数 on each interval $[x_k, x_{k+1}]$ so that the resulting piecewise curve $y = S(x)$ and its first and second derivatives are all continuous on the larger interval $[x_0, x_N]$.
 - ① The continuity of $S'(x)$ means that the graph $y = S(x)$ will not have sharp corners. 无急弯
 - ② The continuity of $S''(x)$ means that the radius of curvature is defined at each point. 曲率不畸变





Cubic Spline Interpolation

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Suitability

分段三次样条曲线:

Definition

Suppose that $\{(x_k, y_k)\}_{k=0}^N$ are $N+1$ points, where $a = x_0 < x_1 < \cdots < x_N = b$. The function $S(x)$ is called a **cubic spline** if there exist N cubic polynomials $S_k(x)$ with coefficients $s_{k,0}, s_{k,1}, s_{k,2}$ and $s_{k,3}$ that satisfy the properties:

- ① $S(x) = S_k(x) = s_{k,0} + s_{k,1}(x-x_k) + s_{k,2}(x-x_k)^2 + s_{k,3}(x-x_k)^3$
for $x \in [x_k, x_{k+1}]$ and $k = 0, 1, \dots, N-1$.
- ② $S(x_k) = y_k$ 通过节点 for $k = 0, 1, \dots, N$.
- ③ $S_k(x_{k+1}) = s_{k+1}(x_{k+1})$ 节点相连 for $k = 0, 1, \dots, N-2$.
- ④ $S'_k(x_{k+1}) = s'_{k+1}(x_{k+1})$ for $k = 0, 1, \dots, N-2$.
- ⑤ $S''_k(x_{k+1}) = s''_{k+1}(x_{k+1})$ for $k = 0, 1, \dots, N-2$.



Existence of Cubic Splines

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存在性:

Is it possible to construct a cubic spline that satisfies all of the five properties?

- Each cubic polynomial $S_k(x)$ has four unknown constants $s_{k,0}$, $s_{k,1}$, $s_{k,2}$ and $s_{k,3}$; hence there are $4N$ coefficients to be determined.

- The data point supply $(N+1)$ conditions (Property 2).
- Properties 3 ~ 5 each supply $(N-1)$ conditions.

$N + 1 + 3 \times (N - 1) = 4N - 2$ conditions are specified.

Two additional conditions necessary for the cubic spline to be solvable come from the boundary conditions for the first/second order derivatives at the end points (x_0, y_0) and (x_N, y_N) .



Boundary Conditions

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Suitability

- Five strategies 五种端点约束条件:

- 1 Clamped cubic spline 紧压样条: specify $S'(x_0), S'(x_n)$ (the “best choice” if the derivatives are known).
- 2 Natural cubic spline (a “relaxed curve”) 自然样条
- 3 Extrapolate $S''(x)$ to the endpoints 外推样条
- 4 $S''(x)$ is constant near endpoints 抛物线终结样条
- 5 Specify $S''(X)$ at each endpoint 端点曲率调整样条

- $4N - 2$ property conditions and 2 endpoint constraints can be used to construct a cubic spline with distinctive properties at the endpoints.

How to construct 怎样构建



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- Since $S(x)$ is piecewise cubic, its second derivative $S''(x)$ is piecewise linear on $[x_0, x_N]$. 二阶导数分段线性

$$S''_k(x) = S''(x_k) \frac{x - x_{k+1}}{x_k - x_{k+1}} + S''(x_{k+1}) \frac{x - x_k}{x_{k+1} - x_k}$$

- Let $m_k = S''(x_k)$, $m_{k+1} = S''(x_{k+1})$ and $h_k = x_{k+1} - x_k$

$$S''_k(x) = \frac{m_k}{h_k}(x_{k+1} - x) + \frac{m_{k+1}}{h_k}(x - x_k)$$

where $x_k \leq x \leq x_{k+1}$, $k = 0, 1, \dots, N-1$



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Suitability

积分两次，引入两个积分常数：

$$S_k(x) = \frac{m_k}{6h_k}(x_{k+1} - x)^3 + \frac{m_{k+1}}{6h_k}(x - x_k)^3 \\ + p_k(x_{k+1} - x) + q_k(x - x_k)$$

$$\begin{cases} y_k &= S_k(x_k) \\ y_{k+1} &= S_k(x_{k+1}) \end{cases} \implies \begin{cases} y_k &= \frac{m_k}{6}h_k^2 + p_k h_k \\ y_{k+1} &= \frac{m_{k+1}}{6}h_k^2 + q_k h_k \end{cases}$$

$$S_k(x) = \frac{m_k}{6h_k}(x_{k+1} - x)^3 + \frac{m_{k+1}}{6h_k}(x - x_k)^3 \\ + \left(\frac{y_k}{h_k} - \frac{m_k h_k}{6}\right)(x_{k+1} - x) + \left(\frac{y_{k+1}}{h_k} - \frac{m_{k+1} h_k}{6}\right)(x - x_k)$$



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求一阶导数：

$$S'_k(x) = -\frac{m_k}{2h_k}(x_{k+1} - x)^2 + \frac{m_{k+1}}{2h_k}(x - x_k)^2 \\ - \left(\frac{y_k}{h_k} - \frac{m_k h_k}{6}\right) + \left(\frac{y_{k+1}}{h_k} - \frac{m_{k+1} h_k}{6}\right)$$

Let $d_k = \frac{y_{k+1} - y_k}{h_k}$, we obtain

$$S'_k(x_k) = -\frac{m_k}{3}h_k - \frac{m_{k+1}}{6}h_k + d_k \\ S'_{k-1}(x_k) = \frac{m_k}{3}h_{k-1} + \frac{m_{k-1}}{6}h_{k-1} + d_{k-1}$$

$$h_{k-1}m_{k-1} + 2(h_{k-1} + h_k)m_k + h_k m_{k+1} = u_k \\ \text{where } u_k = 6(d_k - d_{k-1}) \text{ and } k = 1, \dots, N-1$$



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二阶导数参数方程组:

① $k = 1$, if m_0 is known, then

$$2(h_0 + h_1)m_1 + h_1m_2 = u_1 - h_0m_0$$

② $k = N - 1$, if m_N is known, then

$$h_{N-2}m_{N-2} + 2(h_{N-2} + h_{N-1})m_{N-1} = u_{N-1} - h_{N-1}m_N$$

③ $k = 2, 3, \dots, N - 2$, then

$$h_{k-1}m_{k-1} + 2(h_{k-1} + h_k)m_k + h_km_{k+1} = u_k$$

$$\begin{pmatrix} 2(h_0 + h_1) & h_1 & & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & & \\ & & \ddots & & \\ 0 & & & h_{N-2} & 2(h_{N-2} + h_{N-1}) \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 - h_0m_0 \\ u_2 \\ \vdots \\ u_{N-1} - h_{N-1}m_N \end{pmatrix}$$



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$$h_k = x_{k+1} - x_k, \quad d_k = \frac{y_{k+1} - y_k}{h_k}, \quad u_k = 6(d_k - d_{k-1})$$

$$S_k(x) = s_{k,0} + s_{k,1}(x - x_k) + s_{k,2}(x - x_k)^2 + s_{k,3}(x - x_k)^3$$

$$s_{k,0} = S_k(x_k) = y_k$$

$$s_{k,1} = S'_k(x_k) = -\frac{m_k}{3}h_k - \frac{m_{k+1}}{6}h_k + d_k$$

$$s_{k,2} = \frac{1}{2}S''_k(x_k) = \frac{m_k}{2}$$

$$s_{k,3} = \frac{1}{6}S_k^{(3)}(x_k) = \frac{m_{k+1} - m_k}{6h_k}$$



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Suitability

Specify $S'(x_0)$ and $S'(x_N)$ 紧压样条

$$m_0 = \frac{3}{h_0}(d_0 - S'(x_0)) - \frac{m_1}{2}$$

$$m_N = \frac{3}{h_{N-1}}(S'(x_N) - d_{N-1}) - \frac{m_{N-1}}{2}$$

$$\begin{pmatrix} \frac{3}{2}h_0 + 2h_1 & h_1 & 0 \\ h_1 & 2(h_1 + h_2) & h_2 \\ & & \ddots \\ 0 & h_{N-2} & 2h_{N-2} + \frac{3}{2}h_{N-1} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 - 3(d_0 - S'(x_0)) \\ u_2 \\ \vdots \\ u_{N-1} - 3(S'(x_N) - d_{N-1}) \end{pmatrix}$$

This spline would be useful to a draftsman for drawing a smooth curve through several points.



Natural Spline

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Suitability

Boundary conditions: $S''(x_0) = 0$ and $S''(x_N) = 0$ 自然样条

$$m_0 = 0, \quad m_N = 0$$

$$\begin{pmatrix} 2(h_0 + h_1) & h_1 & & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & & \\ & & \ddots & \ddots & \\ 0 & & & h_{N-2} & 2(h_{N-2} + h_{N-1}) \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{pmatrix}$$

It is useful for fitting curve to experimental data that are significant to several significant digits.



Extrapolated Spline

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Suitability

Use extrapolation from the interior nodes at x_1 and x_2 to determine $S''(x_0)$ and extrapolation from the nodes at x_{N-1} and x_{N-2} to determine $S''(b)$. 外推样条

$$m_0 = m_1 - \frac{h_0(m_2 - m_1)}{h_1}$$

$$m_N = m_{N-1} + \frac{h_{N-1}(m_{N-1} - m_{N-2})}{h_{N-2}}$$

$$\begin{pmatrix} 3h_0 + 2h_1 + \frac{h_0^2}{h_1} & h_1 - \frac{h_0^2}{h_1} & & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & & \\ & & \ddots & & \\ 0 & & & h_{N-2} - \frac{h_{N-1}^2}{h_{N-2}} & 2h_{N-2} + 3h_{N-1} + \frac{h_{N-1}^2}{h_{N-2}} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} =$$



Parabolically Terminated Spline

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Suitability

$S''(x)$ is constant near the endpoints – $S^{(3)}(x) \equiv 0$ on the interval $[x_0, x_1]$ and $S^{(3)}(x) \equiv 0$ on $[x_{N-1}, x_N]$ 抛物线终结样条

$$m_0 = m_1, \quad m_N = m_{N-1}$$

$$\begin{pmatrix} 3h_0 + 2h_1 & h_1 & & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & & \\ & & \ddots & & \\ 0 & & & h_{N-2} & 2h_{N-2} + 3h_{N-1} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{pmatrix}$$

$S^{(3)}(x) \equiv 0$ on the interval $[x_0, x_1]$ forces the cubic to degenerate to a quadratic over $[x_0, x_1]$, and a similar situation occurs over $[x_{N-1}, x_N]$.



Endpoint Curvature-Adjusted Spline

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Suitability

The second derivative boundary conditions $S''(a)$ and $S''(b)$ are specified. 端点曲率调整样条

$$m_0 = S''(x_0), \quad m_N = S''(x_N)$$

$$\begin{pmatrix} 2(h_0 + h_1) & h_1 & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & \\ & & \ddots & \\ 0 & & h_{N-2} & 2(h_{N-2} + h_{N-1}) \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 - h_0 S''(x_0) \\ u_2 \\ \vdots \\ u_{N-1} - h_{N-1} S''(x_N) \end{pmatrix}$$

Imposing values for $S''(a)$ and $S''(b)$ permits the practitioner to adjust the Curvature at each endpoint.



Examples for Cubic Splines

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Suitability

- 1 Find the clamped cubic spline that passes through $(0, 0)$, $(1, 0.5)$, $(2, 2.0)$, and $(3, 1.5)$ with the first derivative boundary conditions $S'(0) = 0.2$ and $S'(3) = -1$.
- 2 Find the natural cubic spline that passes through $(0, 0.0)$, $(1, 0.5)$, $(2, 2.0)$, and $(3, 1.5)$ with the free boundary conditions $S''(x) = 0$ and $S''(3) = 0$.
- 3 Find the extrapolated cubic spline through $(0, 0.0)$, $(1, 0.5)$, $(2, 2.0)$, and $(3, 1.5)$.
- 4 Find the parabolically terminated cubic spline through $(0, 0.0)$, $(1, 0.5)$, $(2, 2.0)$, and $(3, 1.5)$.
- 5 Find the curvature-adjusted cubic spline through $(0, 0.0)$, $(1, 0.5)$, $(2, 2.0)$, and $(3, 1.5)$ with the second derivative boundary conditions $S''(0) = -0.3$ and $S''(3) = 3.3$.



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Suitability

Tridiagonal linear system $HM = U$

$$\begin{pmatrix} b_1 & c_1 & & & 0 \\ a_1 & b_2 & c_2 & & \\ & & \ddots & \ddots & \\ & & & a_{N-3} & b_{N-2} & c_{N-2} \\ 0 & & & a_{N-2} & b_{N-1} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N-1} \end{pmatrix}$$

$$\begin{pmatrix} \frac{3}{2}h_0 + 2h_1 & h_1 & & & 0 \\ h_1 & 2(h_1 + h_2) & h_2 & & \\ & & \ddots & \ddots & \\ & & & h_{N-2} & 2h_{N-2} + \frac{3}{2}h_{N-1} \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_{N-1} \end{pmatrix} = \begin{pmatrix} u_1 - 3(d_0 - S'(x_0)) \\ u_2 \\ \vdots \\ u_{N-1} - 3(S'(x_N) - d_{N-1}) \end{pmatrix}$$

- Strickly diagonally dominant and has a unique solution.



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Suitability

```
function S=csfit(X,Y,dx0,dxn)
N=length(X)-1;
H=diff(X);
D=diff(Y)./H;
A=H(2:N-1);
B=2*(H(1:N-1)+H(2:N));
C=H(2:N);
U=6*diff(D);

% Clamped spline endpoint constraints
B(1)=B(1)-H(1)/2;
U(1)=U(1)-3*(D(1)-dx0);
B(N-1)=B(N-1)-H(N)/2;
U(N-1)=U(N-1)-3*(dxn-D(N));
```



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Suitability

```
for k=2:N-1
    temp=A(k-1)/B(k-1);
    B(k)=B(k)-temp*C(k-1);
    U(k)=U(k)-temp*U(k-1);
end

M(N)=U(N-1)/B(N-1);
for k=N-2:-1:1
    M(k+1)=(U(k)-C(k)*M(k+2))/B(k);
end

% Clamped spline endpoint constraints
M(1)=3*(D(1)-dx0)/H(1)-M(2)/2;
M(N+1)=3*(dxn-D(N))/H(N)-M(N)/2;
```



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```
for k=0:N-1
```

```
    S(k+1,1)=(M(k+2)-M(k+1))/(6*H(k+1));
```

```
    S(k+1,2)=M(k+1)/2;
```

```
    S(k+1,3)=D(k+1)-H(k+1)*(2*M(k+1)+M(k+2))/6;
```

```
    S(k+1,4)=Y(k+1);
```

```
end
```

• Example

```
>> X = [0 1 2 3]; Y = [0 0.5 2.0 1.5]; dx0 = 0.2; dxn = -1;
```

```
>> S = csfit(X, Y, dx0, dxn)
```

```
>> x1 = 0 : .1 : 1;    y1 = polyval(S(1,:), x1 - X(1));
```

```
>> x2 = 1 : .01 : 2; y2 = polyval(S(2,:), x2 - X(2));
```

```
>> x3 = 2 : .01 : 3; y3 = polyval(S(3,:), x3 - X(3));
```

```
>> plot(x1, y1, x2, y2, x3, y3, X, Y, 'o')
```



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三次样条曲线的适合性:

- Among all functions $f(x)$ that are twice continuously differentiable on $[a, b]$ and interpolate a given set of data points $\{(x_k, y_k)\}_{k=0}^N$, the cubic spline has less wiggle.
- Minimum Property of Cubic Splines** 三次样条极小属性.
Assume that $f \in C^2[a, b]$ and $S(x)$ is the unique cubic spline interpolant for $f(x)$ that passes through the points $\{(x_k, y_k)\}_{k=0}^N$ and satisfies the clamped end conditions $S'(a) = f'(a)$ 紧压端点条件 and $S'(b) = f'(b)$. Then

$$\int_a^b (s''(x))^2 dx \leq \int_a^b (f''(x))^2 dx$$



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Minimum Property Proof. 极小属性证明

$$\int_a^b S''(x)(f'(x) - S'(x)) dx \quad \text{应用分部积分}$$

$$= S''(x)(f'(x) - S'(x)) \Big|_{x=a}^{x=b} - \int_a^b S'''(x)(f'(x) - S'(x)) dx$$

$$= 0 - 0 - \int_a^b S'''(x)(f'(x) - S'(x)) dx.$$

Since $S'''(x) = 6s_{k,3}$ on the subinterval $[x_k, x_{k+1}]$, it follows that

$$\int_{x_k}^{x_{k+1}} S'''(x)(f'(x) - S'(x)) dx = 6s_{k,3}(f(x) - S(x)) \Big|_{x=x_k}^{x=x_{k+1}} = 0$$

for $k = 0, 1, \dots, N-1$. Hence $\int_a^b S'''(x)(f'(x) - S'(x)) dx = 0$, and

$$\boxed{\int_a^b S''(x)f'(x) dx = \int_a^b (S''(x))^2 dx.}$$



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Natural Spline

Extrapolated Spline

Parabolically
Terminated Spline

Endpoints
Curvature-Adjusted
Spline

Examples

csfit.m

Suitability

Minimum Property Proof. 极小属性证明

Since $0 \leq (f''(x) - S''(x))^2$, we get the integral relationship:

$$\begin{aligned} 0 &\leq \int_a^b (f''(x) - S''(x))^2 dx \\ &= \int_a^b (f''(x))^2 dx - 2 \int_a^b f''(x) S''(x) dx + \int_a^b (S''(x))^2 dx \\ &= \int_a^b (f''(x))^2 dx - \int_a^b (S''(x))^2 dx \end{aligned}$$

That is

$$\int_a^b (S''(x))^2 dx \leq \int_a^b (f''(x))^2 dx$$